

1. (i) elem nilpotente din inelul $\mathbb{Z}_3 \times \mathbb{Z}_5$

$$\mathbb{Z}_3 \times \mathbb{Z}_5 \simeq \mathbb{Z}_{15}$$

elem nilpotent: (5) $x \in A$ u $x^n = 0$, $x \in \mathbb{Z}_{15}$

$$15 = 3 \cdot 5 \Rightarrow 3 \cdot 5 \mid x \Rightarrow 15 \mid x \Rightarrow A = \{\bar{0}\} \Rightarrow A = \{(\bar{0}, \bar{0})\}$$

(ii) $\mathbb{Z}_{15} \simeq \mathbb{Z}_3 \times \mathbb{Z}_5$

elem idempotent $x^2 = x$, $x \in \mathbb{Z}_3 \times \mathbb{Z}_5$

elem idempotente din $\mathbb{Z}_3 \times \mathbb{Z}_5$ sunt:

$$\begin{cases} (\bar{0}, \bar{0}) = \hat{0} \\ (\bar{1}, \bar{1}) = \hat{1} \\ (\bar{0}, \bar{1}) = \hat{b} \\ (\bar{1}, \bar{0}) = \hat{10} \end{cases} \Rightarrow A = \{\bar{0}, \bar{1}, \hat{b}, \hat{10}\} \text{ elem din } \mathbb{Z}_{15}$$

2. morfismele de inele de la $\mathbb{Z}_{12}$ la $\mathbb{Z}_{19}$

$$\text{fie } \varphi: \mathbb{Z}_{12} \rightarrow \mathbb{Z}_{19}$$

$\varphi: \mathbb{Z}_{12} \rightarrow \mathbb{Z}_{19}$ este morf de inele $\Leftrightarrow$

$$12 \varphi(\bar{1}) = \bar{0}$$

$$\varphi(\bar{1}) = \varphi(\bar{1})^2$$

$$\text{not } \varphi(\bar{1}) = \bar{t}_n, \bar{t}_n \in \mathbb{Z}_{19}$$

$$\begin{cases} \bar{t}_n^2 = \bar{t}_n \\ 12 \bar{t}_n = \bar{0} \end{cases} \Rightarrow \text{căutăm idempotentii din } \mathbb{Z}_{19}$$

$$\mathbb{Z}_{19} \simeq \mathbb{Z}_2 \times \mathbb{Z}_9$$

$$\text{idempotentii sunt: } \begin{cases} (\bar{0}, \bar{0}) = \bar{0} \\ (\bar{1}, \bar{1}) = \bar{1} \\ (\bar{0}, \bar{3}) = \bar{10} \\ (\bar{1}, \bar{0}) = \bar{5} \end{cases} \left\{ \begin{array}{l} \bar{t}_n = \{\bar{0}, \bar{1}, \bar{3}, \bar{10}\} \\ \text{idempotentii} \\ \text{din } \mathbb{Z}_{19} \end{array} \right.$$

die a docci und:

12 $\bar{m} = \bar{0}$ -) da es ist $m = \{\bar{0}, \bar{9}\}$ meng

deci morfisme sunt:

i) $\bar{f}_1(\bar{x}) = \bar{0}$ morf nul

ii) $\bar{f}_2(\bar{x}) = \bar{9}\bar{x}$

b) $\text{Im } \bar{f}_1 = \{\bar{0}\}$

$\text{Im } \bar{f}_2 = \{\bar{0}, \bar{9}\}$

$\text{Ker } \bar{f} = \{x \mid \bar{f}(x) = 0\}$

$\text{Ker } \bar{f}_1 = \mathbb{Z}_{12}$

$\text{Ker } \bar{f}_2 = \{\bar{0}, \bar{2}, \bar{4}, \bar{6}, \bar{8}, \bar{10}\}$